DATA SCIENCE INTERNSHIP

TASK 5

Salary Prediction Using Linear Regression

A beginner-friendly Python machine learning project

Project area	Supervised Machine Learning
Algorithm	Linear Regression
Main tools	Python, pandas, NumPy, Matplotlib, scikit-learn

1. Project Overview

Objective: Build a simple machine learning model that predicts an employee's salary from their years of work experience. This document uses a generated sample dataset so the project can run without downloading a separate file.

Problem type: Regression, because the target value (salary) is a continuous number.

2. Requirements

Use Python 3 with the following libraries: pandas, NumPy, Matplotlib, and scikit-learn. Google Colab is recommended for beginners because these libraries are usually available already.

3. Step-by-Step Instructions

1. Open Google Colab at https://colab.research.google.com/.
2. Create a new notebook.
3. Copy the complete code from the next section into one code cell.
4. Click the Run button or press Shift+Enter.
5. Read the evaluation metrics and inspect the graph.
6. The program saves a dataset and prediction file in the current working folder.

4. Complete Python Code

```
# Task 5: Salary Prediction using Linear Regression

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# 1. Create a sample dataset
np.random.seed(42)
experience = np.round(np.linspace(0.5, 10.4, 100), 1)
noise = np.random.normal(0, 25000, len(experience))
salary = 300000 + (experience * 80000) + noise

data = pd.DataFrame({
    "YearsExperience": experience,
    "Salary": salary.round(0).astype(int)
})

# Save the dataset
data.to_csv("salary_data.csv", index=False)

# 2. Select input (X) and target (y)
X = data[["YearsExperience"]]
y = data["Salary"]

# 3. Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 4. Train the Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)

# 5. Make predictions on test data
y_pred = model.predict(X_test)

# 6. Evaluate the model
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("SALARY PREDICTION - MODEL RESULTS")
print("---------------------------------")
print(f"Training records: {len(X_train)}")
print(f"Testing records: {len(X_test)}")
print(f"Mean Absolute Error (MAE): {mae:,.2f}")
print(f"Root Mean Squared Error (RMSE): {rmse:,.2f}")
```

```
print(f"R-squared (R2): {r2:.3f}")

# 7. Predict salary for a new experience value
new_experience = pd.DataFrame({"YearsExperience": [3.5]})
predicted_salary = model.predict(new_experience)[0]
print(f"Predicted salary for 3.5 years experience: {predicted_salary:,.2f}")

# 8. Plot data and regression line
plt.figure(figsize=(8, 5))
plt.scatter(X, y, label="Sample data")
line_x = pd.DataFrame({
    "YearsExperience": np.linspace(X["YearsExperience"].min(),
                                   X["YearsExperience"].max(), 100)
})
line_y = model.predict(line_x)
plt.plot(line_x, line_y, label="Regression line")
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Salary Prediction using Linear Regression")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig("salary_prediction_plot.png", dpi=150)
plt.show()

# 9. Save test predictions
results = X_test.copy()
results["ActualSalary"] = y_test
results["PredictedSalary"] = y_pred.round(0).astype(int)
results.to_csv("salary_predictions.csv", index=False)

print("Files saved: salary_data.csv, salary_predictions.csv, salary_prediction_plot.png")
```

5. Code Explanation

NumPy: creates sample experience values and random variation. **pandas:** stores the data in a table and saves CSV files. **Matplotlib:** draws the scatter plot and regression line. **scikit-learn:** splits data, trains the model, predicts salaries, and measures performance.

Dataset: 100 example records are created. Salary is generated using an approximate relationship: base salary of 300,000 plus 80,000 for each year of experience, with random variation added. These are synthetic educational data, not real salary statistics.

Train/test split: 80% of the rows train the model and 20% test it on data not used during training.

Linear Regression: learns a straight-line relationship between experience and salary. It estimates the slope and intercept that best fit the training examples.

Evaluation metrics:
• MAE: average absolute difference between actual and predicted salaries; lower is better.
• RMSE: gives larger errors more influence; lower is better.
• R^2: indicates how well predictions explain the variation in test salaries; closer to 1 generally indicates a better fit.

6. Sample Output

Your exact numbers may differ if you modify the code or dataset. With the code as written, output will look like this:

```
SALARY PREDICTION - MODEL RESULTS
--------------------------------
Training records: 80
Testing records: 20
Mean Absolute Error (MAE): [value printed by Python]
Root Mean Squared Error (RMSE): [value printed by Python]
R-squared (R2): [value printed by Python]
Predicted salary for 3.5 years experience: [value printed by Python]
Files saved: salary_data.csv, salary_predictions.csv, salary_prediction_plot.png
```

The program also displays a chart with the sample salary points and the fitted regression line. Use the real printed metric values from your own run in your final submission.

7. Files Produced

• **salary_data.csv** — generated sample data.
• **salary_predictions.csv** — actual and predicted salaries for test records.
• **salary_prediction_plot.png** — visualization of the data and fitted line.

8. Project Report

Title: Salary Prediction Using Linear Regression

Introduction: Salary prediction is a regression problem in which a model estimates salary using relevant information. In this beginner project, years of experience is used as the input feature.

Objective: To understand the basic machine learning workflow: preparing data, splitting it into training and testing sets, training a regression model, evaluating predictions, and visualizing results.

Methodology: A synthetic dataset of 100 records is generated with years of experience and corresponding example salaries. The data is divided into training and testing subsets. A Linear Regression model is trained using the training subset, then predictions are compared with test values using MAE, RMSE, and R^2.

Result: The program produces a predicted salary for a person with 3.5 years of experience, reports model evaluation metrics, creates a graph, and saves the data and predictions as CSV files. Insert the exact values from your run if your internship report requires numerical results.

Conclusion: This project demonstrates how Linear Regression can model a simple relationship between work experience and salary. The results are for learning only because the dataset is synthetic. Real salary estimates

would require reliable real-world data and additional features such as role, location, education, and industry.

9. Troubleshooting

• If a library import fails, run `!pip install numpy pandas matplotlib scikit-learn` in a separate Google Colab cell, then rerun the code.
• If the chart does not appear, make sure the cell completed without an error.
• If you need to submit the output, run the code and take screenshots of the printed metrics and graph.
• Do not report sample metric values that you did not obtain from your own run.

End of Task 5 document